

parity between assets A and B, the impermanent loss disappears. If the goal for liquidity held in the contract is profit, any fees charged must exceed the amount of the impermanent loss.

Impermanent loss occurs for any shift in price and liquidity because the contract is structured to sell the appreciating asset and to buy the depreciating asset. An important feature of impermanent loss is *path independence*. In our example, it is irrelevant whether 1 or 100 traders consumed all the liquidity. The final exchange rate and contract asset ratios yield the same impermanent loss regardless of the number of trades or the direction of the trades. Because of path independence, impermanent loss is minimized on trading pairs that have correlated prices (*mean-reverting pairs*). Thus, stablecoin trading pairs are particularly attractive for AMMs.

COLLATERALIZED LOANS

Debt and lending are perhaps the most important financial mechanisms that exist in DeFi and, more generally, in traditional finance. On one hand, these mechanisms are a powerful tool for efficiently allocating capital, increasing return-bearing risk exposure, and expanding economic growth. On the other hand, excess debt in the system can cause instability, potentially leading to large economic and market contractions. These benefits and risks are amplified in DeFi because the counterparties share an adversarial and integrated environment. Platforms are increasingly